

Paper Review: A Simple Sequentially Rejective Multiple Test Procedure (Holm, *Scand. J. Statist.*, 1979)

1. Problem Statement

When researchers perform multiple statistical tests at the same time—such as comparing several different machine learning models, evaluating multiple clinical criteria, or testing various patient outcomes—the chance of making a false positive error (Type I error) increases drastically. If you test 20 hypotheses at a standard significance level of 0.05, you are almost guaranteed to find at least one "statistically significant" result purely by random chance. Traditionally, procedures like the Bonferroni correction were used to control this error rate by making the threshold very strict. However, the Bonferroni correction is extremely conservative and lacks statistical power, meaning it often misses truly important discoveries. The problem Holm solved was how to control the family-wise error rate across multiple hypothesis tests without losing too much statistical power.

2. Body Summary

In this classic 1979 paper, Sture Holm introduced a simple, elegant, and universally applicable sequentially rejective multiple test procedure (commonly known today as the **Bonferroni–Holm method** or simply **Holm's step–down procedure**). Instead of testing all hypotheses against a single rigid threshold, Holm's method orders the p-values from smallest to largest and tests them step–by–step. If the smallest p-value is significant under an adjusted threshold, it gets

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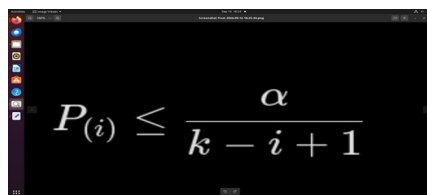
rejected, and the test moves to the next one with a slightly more lenient threshold. This simple algorithmic adjustment strictly controls the family-wise error rate at level α under any dependency structure between the tests, while being significantly more powerful than the traditional Bonferroni correction.

3. Core Architecture & Methodology Used

The methodology relies on step-down statistical hypothesis testing:

- **P-Value Sorting:** Take “k” null hypotheses and calculate their respective p-values starting from P_1 till P_k . Sort them in ascending order such that $P(1), P(2), \dots, P(k)$

- **Step-Down Comparison:** Compare each sorted p-value against an adjusted significance threshold:


$$P_{(i)} \leq \frac{\alpha}{k - i + 1}$$

- **Sequential Rejection Rule:** Start with the smallest p-value $i = 1$. If $P(1)$ is less than or equal to α divided by “k”, reject the first null hypothesis and move to the next. Continue testing sequentially until a p-value fails the condition, at which point you stop and accept all remaining hypotheses.

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4. Logical Reasoning & Conclusion

1. **Logical Reasoning:** This paper is crucial when you evaluate your MTP clinical trial matching models. If you are comparing multiple configurations (e.g., LLM-only vs. RAG vs. your new hybrid verification framework) across various evaluation metrics or patient subsets, you cannot just look at raw score differences without accounting for multiple testing correction. Holm's method ensures that your performance improvements or error rate reductions are statistically robust and not just flukes.
2. **Conclusion:** Holm (1979) provided a foundational statistical tool that balances rigorous error control with high statistical power. It remains a gold standard for multiple comparisons across statistics, data science, and clinical research evaluations.